

Landslides (2025) 22:2701–2713
 DOI 10.1007/s10346-025-02525-8
 Received: 6 September 2024
 Accepted: 11 April 2025
 Published online: 3 May 2025
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Analysing dynamics of the 2000 Yigong landslide in the Tibetan plateau using seismic observations

Abstract Climate change and urbanisation increase the risk of landslides, particularly mass movements with high mobility that can travel long distances and cause significant casualties and economic losses. Despite their impact, the mechanisms controlling long-runout landslides remain poorly understood due to their complexity. This study investigates the catastrophic 2000 Yigong landslide on the Tibetan plateau, using seismic signals to invert for the force–time function during its movement and evaluate changes in the basal friction coefficient. We identified five distinct stages in the event’s dynamics, beginning with two rock collapses in the source zone during the acceleration phase. In the entrainment phase, the mass attained its maximum velocity and forces when it impacted the Zhamu Creek. Notably, the equivalent friction coefficient dropped from 0.58 to 0.06, indicating a transition from rock collapse to debris flow. Our velocity–friction analysis aligns with flash heating theory, suggesting that high-speed sliding between rough surfaces generated heat, which partially melted the glacier and introduced fluid into the sliding process, resulting in enhanced mobility and facilitated long-runout distances down the Zhamu Creek. This study highlights the value of seismic signals in understanding the physical mechanisms behind catastrophic landslides and addressing the challenges of monitoring these geohazards in remote regions.

Keywords Environmental seismology · Yigong landslide · Seismic inversion · Landslide dynamics · Basal friction

Introduction

Landslides are a global hazard on unstable slopes. Intensifying climate conditions, such as extreme rainfall and rising temperatures, are increasing pore pressure in soils and rocks, leading to more frequent and complex landslides (Huggel et al. 2012; Handwerger et al. 2019; Morino et al. 2021). These landslides pose threats to lives and infrastructure in mountainous regions. Their long-term cascading effects, like dammed lakes and outburst floods, further amplify risks and uncertainties downstream over large areas (Zhang et al. 2024a, b). High-mobility ground slumping is an increasingly alarming surface geohazard, as these long-runout landslides (Guo et al. 2023) travel long horizontal distances with great drop heights and large volumes, often resulting in catastrophic, widespread, and high-velocity impacts. The steep terrain and harsh conditions of the Tibetan Plateau make it one of the most landslide-prone regions on Earth, where some of the largest-known valley-blocking landslides have occurred. For example, the Sedongpu debris flow and Baige landslide in 2018 blocked the Jinsha River and the Yarlung Zangbo River, respectively, with the subsequent dam breach producing

catastrophic outburst floods (Gao et al. 2023b; Zhang et al. 2024b), which projected a hazard for hundreds of kilometres downstream. Understanding the dynamics and evolution of historic landslides is crucial for reducing uncertainties and developing improved hazard management strategies, such as future landslide forecasting, monitoring, and warning systems. However, there is insufficient near-real-time data to detect the disasters and verify the various hypotheses proposed to quantify the mechanisms driving these long-runout landslides, such as friction weakening, flash heating, and changes in volumetric water content (Singer et al. 2012; Lucas et al. 2014; Goren and Aharonov 2007).

The mobility of a landslide refers to the ease with which its materials can be eroded and transported during mass wasting (Pudasaini and Krautblatter 2021). This mobility plays a crucial role in determining the runout distance. Friction along the basal sliding plane governs the resistance to motion and the ease of movement, it is often quantified by a friction coefficient, where higher values indicate greater resistance and lower mobility, and vice versa (Byerlee 1978). Constraining the friction coefficient is essential for studying landslides, as it reflects the physical properties and processes of the mass, such as volume, pore fluid content, and slope angle, which influence the initiation mechanisms, kinetic and kinematic variations, and the state of the mass (Brodsky et al. 2003; Lucas et al. 2014; Handwerger et al. 2016; Yamada et al. 2018). Numerous studies have demonstrated that friction directly affects the velocity and distance a landslide travels, and the forces it exerts on surrounding structures and terrain (Castelli et al. 2009). Various methods have been proposed to estimate the basal friction coefficient, which include relating it to the drop height and impact features (Singer et al. 2012; Zhang and Yin 2013; Lucas et al. 2014), as well as laboratory experiments exploring potential weakening mechanisms (Wang et al. 2017; Deng et al. 2020). However, the complexity in studying friction and the lack of approaches to analyse high-resolution real-time data have hindered our understanding of the initiation mechanisms and evolution dynamics of landslides. Additionally, more research is needed to clarify how variations in the surrounding environment govern the state and friction throughout a slide.

Seismic sensors record the seismic energy radiated by landslides during runout, allowing estimation of basal friction evolution (Brodsky et al. 2003; Allstadt 2013; Lin 2015; Yamada et al. 2018). Analysing these seismic observations and other related geophysical data allows the investigation of the kinematics of mass wasting events at high temporal resolution (Yamada et al. 2012; Allstadt 2013; Ekström and Stark 2013; Lin 2015; Zhang et al. 2019), because the structural characteristics of the sliding materials and

the spatiotemporal extent of the landslide significantly influence the amplitude and frequency of the seismic waveforms (Lin et al. 2010; Yamada et al. 2012; Allstadt 2013; Hibert et al. 2014; Zhang et al. 2019). Large landslides move coherently as blocks and generate long-period signals detectable globally, as the bulk acceleration and deceleration of the moving mass produces forces down the slope (Lin et al. 2010; Yamada et al. 2012; Allstadt 2013; Hibert et al. 2014; Zhang et al. 2019). In contrast, high-frequency seismic energy has been inferred for incoherent, fine-grained sediment motions at a local scale (Suriñach et al. 2005; Deparis et al. 2008; Allstadt 2013; Zhang et al. 2020). The envelope of these short-period signals has also been shown to correlate well with the momentum of large landslides (Hibert et al. 2017). Previous studies have established that long-period signals reflect large-scale bulk momentum variations, while high-frequency signals are linked to smaller-scale momentum exchanges (Kanamori and Given 1982; Fukao 1995; Ekström and Stark 2013; Hibert et al. 2014). For example, a similar approach was well demonstrated by Allstadt (2013) for the 2010 Mount Meager landslide in British Columbia, Canada. Therefore, seismic signals allow for understanding the forces exerted by landslides at different spatial scales during runout, which is crucial for explaining the underlying physical mechanisms.

This study aims to understand the dynamic evolution of the 2000 Yigong landslide, particularly the evolution of basal friction, using high-resolution seismic signals along with satellite imagery and field data. We inverted the long-period seismic signals to obtain the time series of forces exerted by the landslide's centre of mass. From this, we calculated the acceleration, velocity, and displacement, which provided insights into the equivalent basal friction coefficient of the centre of mass. By aligning these modelled forces and trajectory results with auxiliary observational data, such as terrain features and satellite imagery, we gained valuable insights into the source properties, dynamics, and friction evolution of this large, long-runout landslide.

Data

The 2000 Yigong landslide

The Yigong landslide occurred on April 9, 2000 at around 20:00 local time (UTC + 8) at 30.23°N, 94.98°E in southeastern Tibet, China. A massive glacier and rock debris flow lasting ~ 10 min was reported by Yigong Village residents (Zhou et al. 2001; Shang et al. 2003; Delaney and Evans 2015). The landslide initiated from the southwest-facing slope of the Zhamu Creek's rear edge (Fig. 1d), with tension cracks observed at the rear of the slope (Shang et al. 2003; Xin et al. 2022). The collapsed mass was estimated at $1.8 \times 1.2 \times 1.26$ km (max. $L \times W \times D$) based on measurements from satellite images and elevation data using a GIS tool, totalling 3×10^8 m³ (Shang et al. 2003; Ekström and Stark 2013). The site is in the Tibetan alpine environment (Fig. 1a), with glacier ice in the source area and vegetated banks along the Zhamu Creek, which feeds into the Yigong-Tsangpo River. The deposits blocked the Yigong-Tsangpo River, forming a dammed lake for 62 days before a catastrophic breach and outburst flood on June 10 (Fig. 1b). Prior to the 2000 event, five other landslides had dammed the Yigong-Tsangpo River in this area around 3500 BC, 1300 BC, 1000 BC, 600 BC, and 1900, based on field surveys and dating analyses (Guo et al. 2020).

The Zhamu Creek is a 10-km long V-shaped valley, with a maximum elevation of ~ 5520 m a.s.l. The collapsed source sits at 5260 m a.s.l., and the slumped mass was deposited between 2143 and 2376 m a.s.l., with the mean at 2296 m based on GMRT DEM topographic data. The runout distance of the mass centre was 8.79 km, resulting in a Heim's ratio of 0.230 for the 2000 Yigong landslide. The slope of the Zhamu Creek ranges from 40 to 60° (Guo et al. 2023). The geology of Yigong is complex and heavily influenced by tectonic activity from the Himalayan Orogeny, resulting in high-intensity metamorphism and deformations (Shang et al. 2003; Delaney and Evans 2015; Dai et al. 2021; Guo et al. 2023). The bedrock consists of a variety of metamorphic and sedimentary rocks, including Himalayan granite in the source zone, as well as marble, limestone, slate, and siltstone along the gorge. Overlying the Zhamu Creek is a layer of unconsolidated scree from the 1900 landslide (Guo et al. 2020). The Zhamu Creek lies along the northeast-southwest trending Yigong-Lulang Fault, which intersects the Jiali Fault at an orthogonal angle (Lee et al. 2003; Shang et al. 2003) (Fig. 1c). These faults are active, and seismicity is frequent in the region (Mukhopadhyay and Dasgupta 2015). The sinistral motion of the Yigong-Lulang Fault creates a flower structure with subsidiary faults and extensional fractures bounding the creek's rear edge (Shang et al. 2003; Xin et al. 2022), predisposing the area to slope failures and forming the landslide source zone and forming the source area.

Seismic observations

The 2000 Yigong landslide generated seismic signals detected by over 36 stations up to 3600 km away (Fig. 2a), with an enormous energy dissipation that was observed teleseismically (Ekström and Stark 2013). The seismic onset was recorded at 12:01:02 UTC on the nearest broadband station, LSA, located 378.4 km from the source in Lhasa, Tibet (IC network) (Fig. 2c and d). As a surficial mass movement, the Yigong landslide signal exhibited prominent surface wave features without clear P and S wave arrivals, gradually building up a lens-shaped amplitude characteristic (Suriñach et al. 2005; Deparis et al. 2008; Allstadt 2013; Ekström and Stark 2013; Hibert et al. 2014). The seismic response extracted from station LSA began with a series of low-amplitude signals lasting 52 s (Fig. 3a), before intensifying with multiple peaks between $t = 42$ s and $t = 75$ s, then gradually fading over the next 400 s. Spectrograms showed a triangular signature in the frequency domain, with effective frequencies concentrated below 0.4 Hz (Fig. 2e), and a long duration of approximately 300 s (Ekström and Stark 2013; Zhang et al. 2019). These seismic features were consistent across the regional seismic stations (Fig. 2a and b).

Data selection

In addition to the teleseismic detection of the Yigong landslide (Ekström and Stark 2013), we obtained data from selected seismic stations located 300–1000 km from the Zhamu Creek, covering north, east, south, and west directions. This station selection criterion ensures the distance is large enough to consider the landslide as a point source, while the generated signals are not overly attenuated (Toney and Allstadt 2021). The broad azimuthal coverage also helps minimise potential bias in seismic wave propagation during

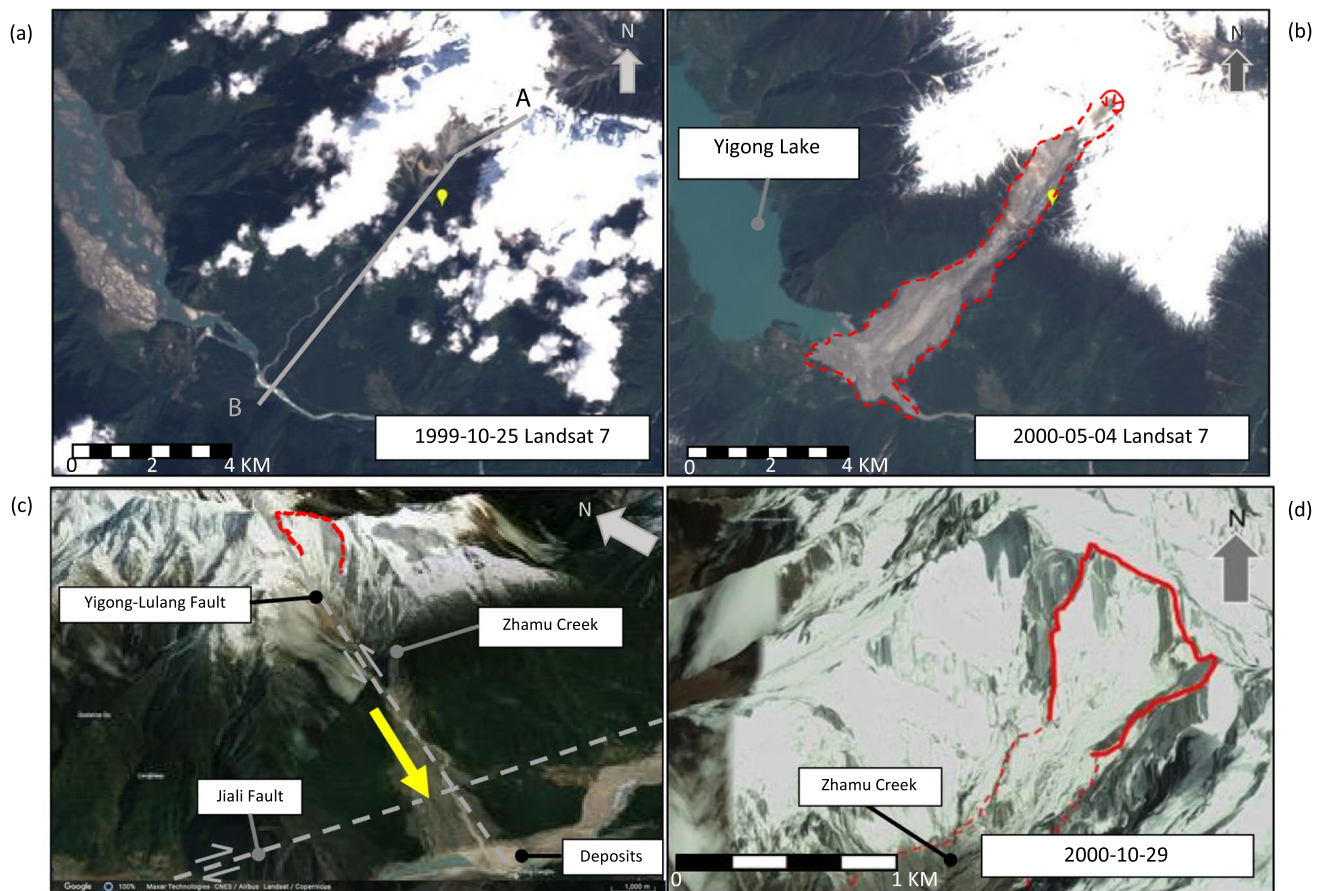


Fig. 1 Geographical and geological overview of Yigong. **a** Satellite image from before the 2000 Yigong landslide. **b** Satellite image less than a month after the 2000 event showing the scar (outlined in dashed red line) with sediments deposited down the creek and damming the Yigong-Tsangpo River. **c** Major structural features in Yigong. Source area of the 2000 landslide in red, regional faults in grey, and direction of slump marked by yellow arrow (after Dai et al. 2021; extracted from Google Earth). **d** A close-up view of the crown of the 2000 Yigong landslide captured by a satellite image sourced from Landsat/Copernicus on October 29, 2000. The solid red line outlines the V-shaped scar in the source region, which features an angular cliff face at the top of Zhamu Creek (5260 m a.s.l.). This corresponds to the two mass collapses observed in the field (Xu et al. 2012; Xin et al. 2022), with seismic interpretations in this study indicating movement first from the east of the cliff and then from the north. The dashed red lines indicate the path along which the mass slid down Zhamu Creek, showing lineations created by the dragging of snow and soil. The satellite image was last accessed via Google Earth Pro on August 29, 2024

inversion (Allstadt 2013; Toney and Allstadt 2021). We then selected seismic data based on the signal-to-noise ratio of the radial, transverse, and vertical components at each station (Table S1).

Method

We adopted the methodology outlined by Allstadt (2013) and processed the inversion using the Python-based package, *lsforce*, developed by Toney and Allstadt (2021). For a simplified model of a sliding block mass on an incline, the net force (F) is an equal reactive force experienced by the sliding mass at the slope surface in an opposing direction of the sliding force (F_s), which arises from the difference between the gravitational force, the counteracting friction, and the centripetal forces in Eq. (1). F can also be described as a single-force mechanism within the long-wavelength limit (Kanamori and Given 1982; Takei and Kumazawa 1994; Fukao 1995; Allstadt 2013; Ekström and Stark 2013; Li et al. 2019). Topographic variations and lateral displacements during the slide are trivial as

they have little influence on the long-period seismic signals when compared to the distances from the landslide source to the seismic stations (Allstadt 2013; Toney and Allstadt 2021).

$$F = -F_s = -ma = -mg(\sin\theta - \mu\cos\theta) \quad (1)$$

where m is a constant mass of sliding block, a corresponds to the acceleration of mass, gravitational acceleration $g = 9.81 \text{ ms}^{-2}$, θ is the slope angle, and μ represents the equivalent dynamic coefficient of friction that accounts for basal pore fluid pressure.

Inversion setup

We invert for the force–time function from long-period seismic signals generated by the landslide and recorded by the seismometers (Table S1). The displacement seismograms ($d(t)$) at each station are obtained by convolving ($\ast$) the force–time function ($F(t)$) with the Green's functions ($g(t)$) at the source-to-station distance:

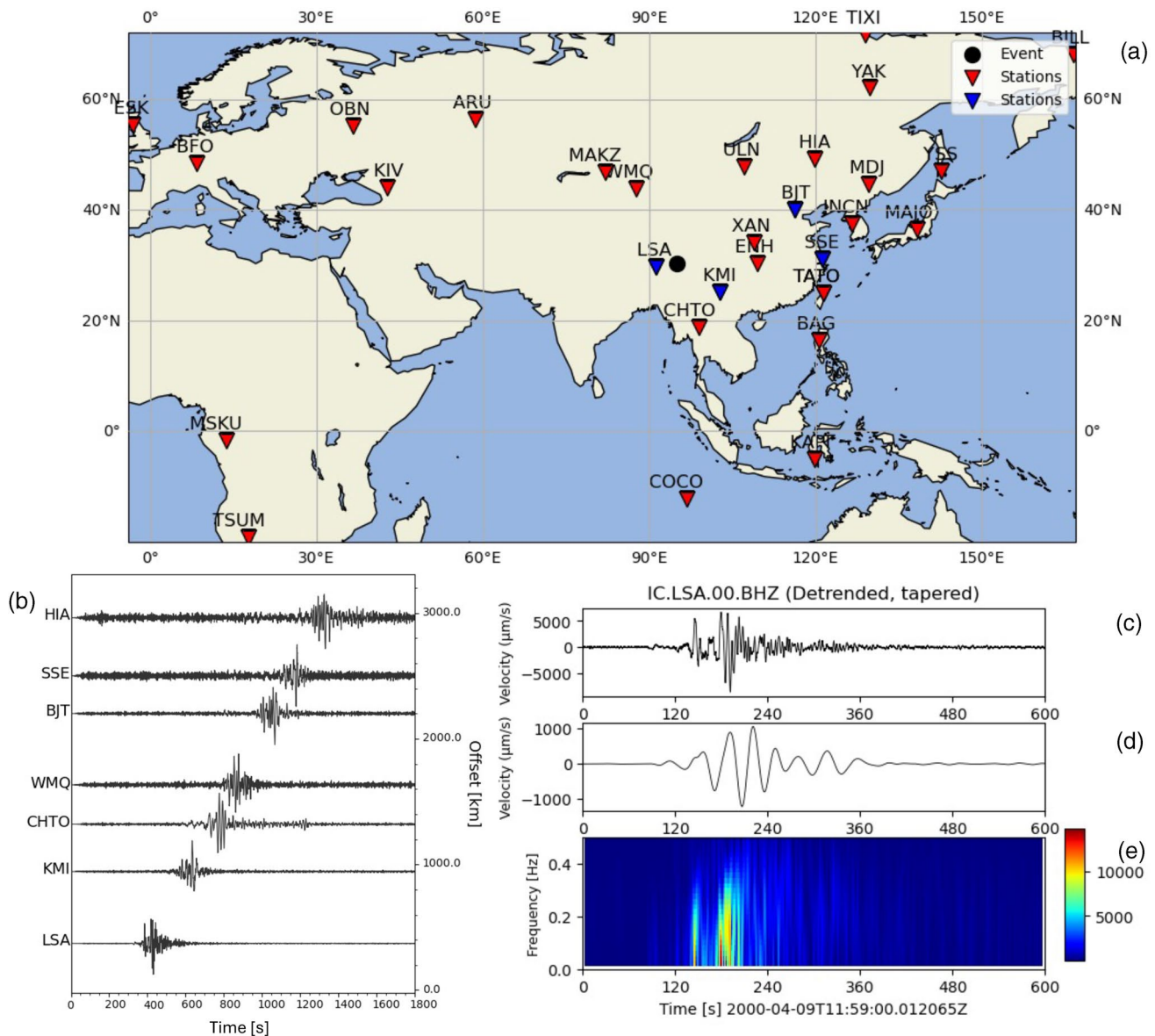


Fig. 2 **a** Regional map showing the location of Yigong and selected stations (in red) with records of seismic signals generated by the 2000 landslide. Stations in blue are used in this study for seismic inversion. **b** Seismic signals of the vertical component generated by the 2000 Yigong landslide were recorded at stations within 3000 km from the source. **c** Zoom-in vertical component seismic data of the 2000 landslide from the nearest station, IC. LSA, unfiltered seismogram. **d** Same seismic waveform is bandpass filtered at 30–150 s. **e** Spectrogram of seismic signals with a colour bar showing the amplitude of signals

$$d(t) = F(t) * g(t) \quad (2)$$

We model the observed waveforms as synthetic seismic records, which capture the effects of source $F(t)$ and wave propagation $g(t)$ through the Earth at each station as the three-dimensional displacements over time. Green's functions mathematically describe the response of the three-dimensional displacements induced by a unit impulse force in vertical (V) and horizontal (H) directions. The resultant three-dimensional force–time series is expressed in north (N), east (E), and vertical (Z) directions, $F(t) = [f_N(t), f_E(t), f_Z(t)]$, which corresponds to the ground displacements in the vertical (Z), radial (R), and transverse (T) components accounting for the

azimuth ($\varnothing$) from source to station clockwise from north that accounts for the radiation pattern of seismic waves. A positive vertical force component indicates downward motion of mass, while the north and east forces describe the direction of slide (Allstadt 2013).

The Green's functions for all stations are precomputed and retrieved from the Incorporated Research Institutions for Seismology (IRIS) Synthetics Engine web service (Syngine, Krischer et al. 2017) using the IASP91 Earth velocity model (Kennett 1991). The start time of the event is defined as 11:58:56 UTC from the seismic data (Fig. 2c–e) and feedback from the computed trajectory. The observed and synthetic data at all stations are resampled at 0.55 Hz and bandpass filtered at two frequency ranges: 10–150 and

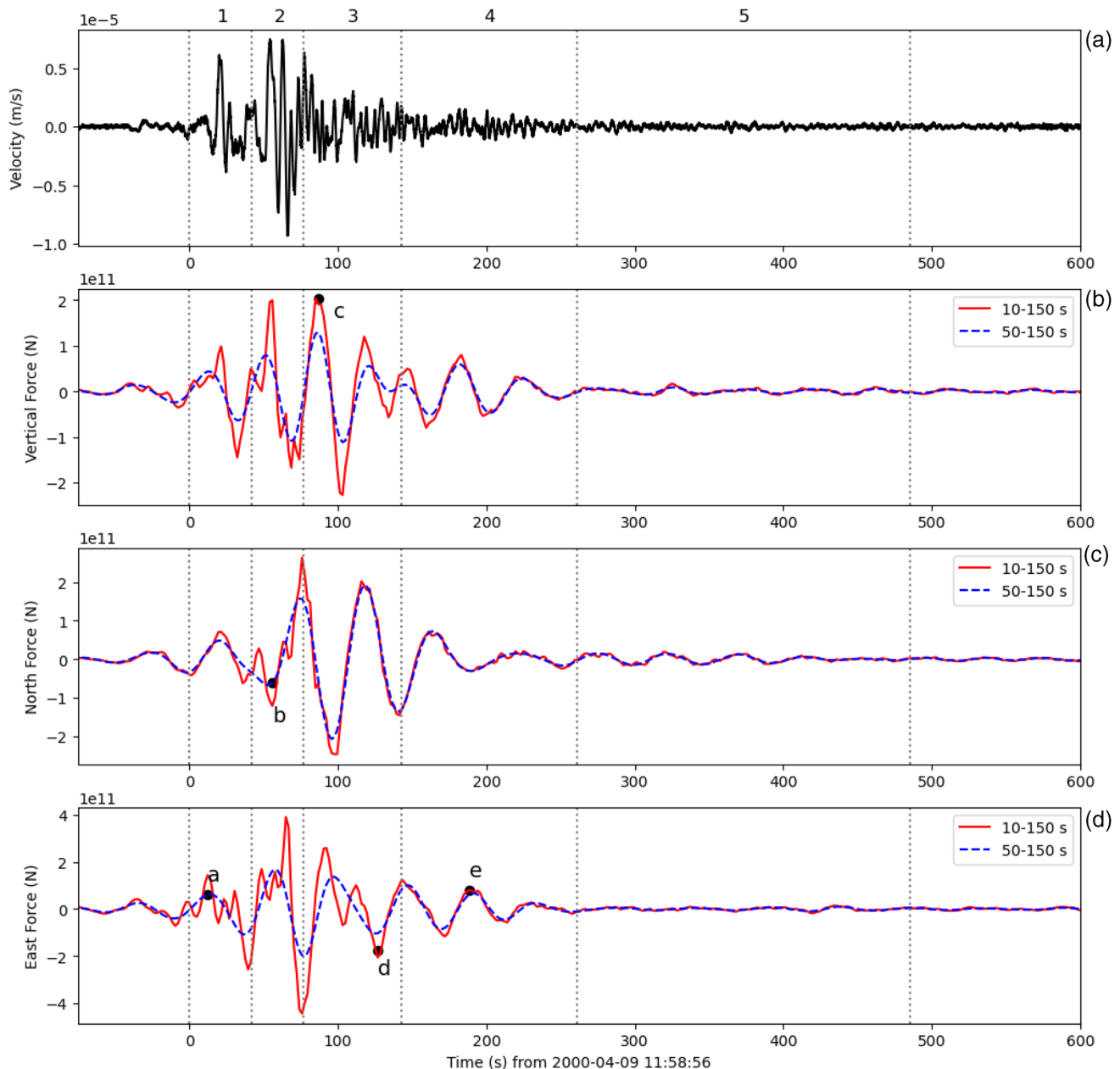


Fig. 3 Seismogram and force–time functions with stages 1 to 5 labelled. **a** Seismogram recorded at IC.LSA; **b** vertical force; **c** horizontal force (north); and **d** horizontal force (east). Points a–e are the features of each stage along the runout shown in Fig. 4a. The forces inverted from 50- to 150-s seismic signals correspond to the landslide dynamics of coherent bulk mass collapses during the acceleration phase, whereas the forces inverted from 10- to 150- s seismic signals correspond to finer scale complexities and minor disturbances at the basal sliding plane during debris flow

50–150 s. The 10- to 150-s bandpass filtered seismic signal (Fig. S1a) highlights the less cohesive, localised disturbances and debris flow complexities during the entrainment and deposition phases of the landslide (Allstadt 2013; Zhang et al. 2020). The 50- to 150-s range (Fig. S1b) captures the coherent bulk mass collapses during the acceleration phase, as these surface waves generated by the landslide are characterised by long-period, low-frequency Rayleigh and Love waves (Aki and Richards 1980).

Trajectory

Using the best-fitted three-component force–time functions from seismic inversion with the assumptions stated, we calculate the centripetal acceleration of the landslide by rearranging Eq. (1), then integrate the time-varying acceleration once to yield the velocity and twice to compute the displacement of the slide.

$$u(t) = \int v(t)dt = \iint a(t)dt = \frac{1}{m} \iint F(t)dt \quad (3)$$

For more in-depth kinematic analyses, we resolve for the landslide's runout path and direction. With georeferenced satellite imagery but a lack of instantaneous topographic data from the digital elevation model (DEM) in the Yigong area, we validate the landslide trajectory using a Landsat 7 satellite image captured on May 4, 2000, which gives a measured runout distance of the landslide centre of mass as 8.79 km. Additionally, we define the event duration as 485 s, which is observed from the inverted force–time functions (Fig. 4) and seismic records (Fig. 2c).

We further calculate the equivalent coefficient of friction over time,

$$\mu'(t) = \left(\sin\theta - \frac{a(t)}{g} \right) \cos^{-1}\theta \quad (4)$$

Instead of calculating an average coefficient as in previous studies (Brodsky et al. 2003; Allstadt 2013; Zhang et al. 2019), we incorporate time-varying acceleration derived from seismic inversion and the corresponding slope angle interval to estimate the evolution of basal friction. The typical friction coefficient range for Yigong lithologies is between 0.2 and 0.8 (Byerlee 1978), although the steep slope angles, 8° to 53°, may slightly overestimate the results due to the sensitivity of the $\cos^{-1}\theta$, yielding coefficients ranging between −1 and 1, and potentially exceeding 1.

Results

Inversion validation

The inversion analysis achieved a 72% variance reduction, indicating a promising consistency between the observed and synthetic data (Fig. S2), suggesting reliable results from the inversion (Toney and Allstadt 2021). Furthermore, the force–time series exhibited a consistent pattern across the 10- to 150- and 50- to 150-s filter bands (Fig. 3b–d), reinforcing the robustness of the interpretation. Additionally, variations at both higher and lower frequencies reflect the intricate dynamics of the landslide (Kanamori and Given 1982; Fukao 1995; Ekström and Stark 2013; Hibert et al. 2014). We also validate the trajectory of the centre of mass by overlaying the time-lapse horizontal displacements with satellite imagery (Fig. 4a). The computed trajectory closely aligns with the crown at source, the scar and topographic depression along Zhamu creek, and the deposits at the Yigong River. The time-lapse analysis tracks the position of the sliding mass throughout the event with time stamps (Fig. 4a), where we identified five points, *a* to *e*, that correspond to local maxima in the force history and the potential impact points (Fig. 4a). Based on the force history profile and the inferred dynamic parameters (Figs. 3b–d and 4a), we identified five distinct stages of the landslide.

The dynamic parameters of the 2000 Yigong landslide

By integrating seismic data and force–time analyses, we determined the landslide initiated at 23:58:56 and lasted 485 s until seismic signals returned to background levels. From the onset ($t = 0$ s) to $t = 77$ s, the force–time functions exhibited two prominent pulses of

increasing amplitudes corresponding to Stages 1 and 2 (Fig. 3b–d), consistent in both the 10- to 150- and 50- to 150-s band filtered analysis. However, these pulses are characterised by frequent sub-peaks in the 10–150 s force–time function, which are most significant in the vertical and east components (Fig. 3b and d). In addition, the inferred sliding velocity increases from an average of 15 to 30 ms^{-1} during these stages (Fig. 5a). Our results also show two phases of acceleration with two distinct rates of change in the displacement–time graph, at 1.19 ms^{-2} in stage 1 and 2.13 ms^{-2} in Stage 2 (Fig. 5c). We compared the theoretical acceleration using the equation of motion with that we inverted from seismic (Fig. 5c). At $t = 77$ s, peak force magnitudes reached 1.92×10^{11} N (vertical), 2.37×10^{11} N (north), and -4.31×10^{11} N (east), at its maximum sliding velocity of 43.7 ms^{-1} . The mean amplitudes subsequently decreased from 2.44×10^{11} N to 0.69×10^{11} N and then to 0.12×10^{11} N for Stages 3, 4, and 5, where the velocity then stabilised at approximately 35 ms^{-1} before gradually decelerating.

However, the equivalent friction coefficient (Fig. 5b) displayed an inverse relationship with the sliding velocity (Singer et al. 2012; Lucas et al. 2014). Consistent with laboratory measurements for granite (Byerlee 1978), the friction coefficient at 0.56 at the source and fluctuated between 1.24 and 0.5, with a mean coefficient of 0.91 in Stage 1. In Stage 2, it initially remained within the expected range but then decreased to a mean of 0.38. The coefficient sharply dropped to 0.02 during the transition from Stage 2 to Stage 3 at 1.51 km ($t = 77$ s) before gradually stabilising between 0.2 and 0.4 with an average of 0.28. By Stage 5, the coefficient remained steady at 0.32.

The five-stage process of the evolution of the 2000 Yigong landslide

Precursors

Prior to the main collapse, the seismic data is characterised by 52 s of high signal-to-noise ratio (34.26) in the long-period signals. The force–time functions show that the sliding mass exerted a weak, continuous downward force during this time (Fig. 3b–d), potentially representing a precursory mass movement leading to the 2000 Yigong landslide. This motion may reflect early disintegration or gradual collapse of rock fragments during creep-tension-shear processes (Xin et al. 2022; Gao et al. 2023a), as we observed sub-peaks in the vertical force function of the 10- to 150-s band filter analysis (Allstadt 2013). However, our force analyses, which are based on long-period seismic signals, pertain only to the centre of the coherent bulk mass. The distances travelled by these fragments were insignificant and beyond seismic resolution. Attempts to calculate the trajectory, including the 52 s of precursory motion, did not align with the observed runout path on satellite imagery. Consequently, we have excluded this portion of the motion and dynamics for the 2000 Yigong landslide.

Initiation: Stages 1 and 2

In Stage 1, $t = 0$ s to $t = 42$ s, a mass collapsed from 5260 m a.s.l. at the source and first travelled 0.56 km towards south-southwest down a 46.15° slope at 15.22 ms^{-1} with an average acceleration of 1.19 ms^{-2} . The slide was initiated at 11:58:56 as indicated by the onset of

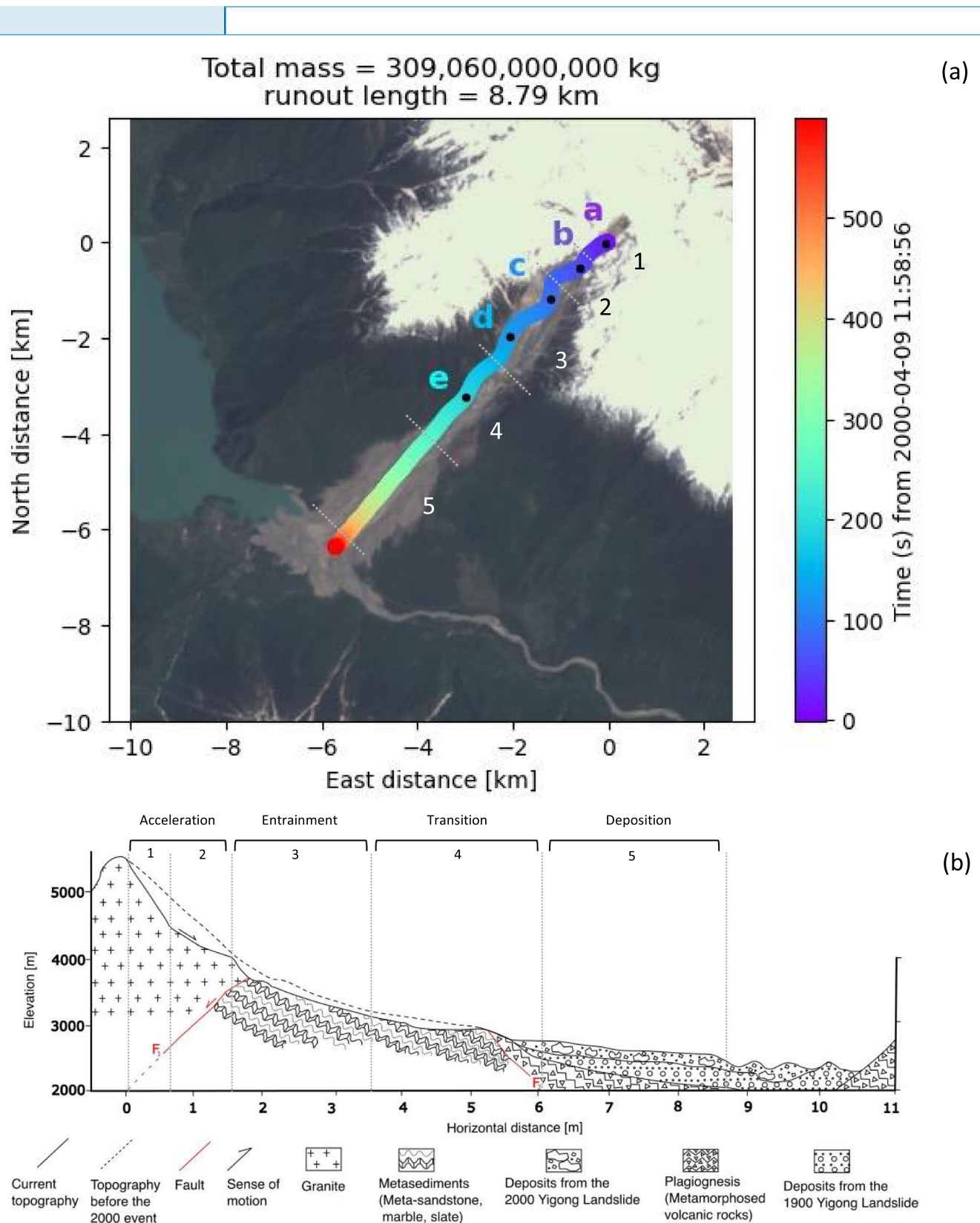


Fig. 4 **a** Horizontal trajectory of the centre-of-mass over the duration of the landslide from seismic inversion (10–150 s) in this study. The displacement–time model cross-references with a satellite image taken from Landsat 7 on May 4, 2000 for spatial validation. Points a–e are the features of each stage (labelled as stages 1–5) along the runout showing in Fig. 3. **b** Topographic profile along the runout with reference to the dynamic stages and processes we defined based on our results from seismic inversion (modified after Xu et al. 2012; Guo et al. 2023)

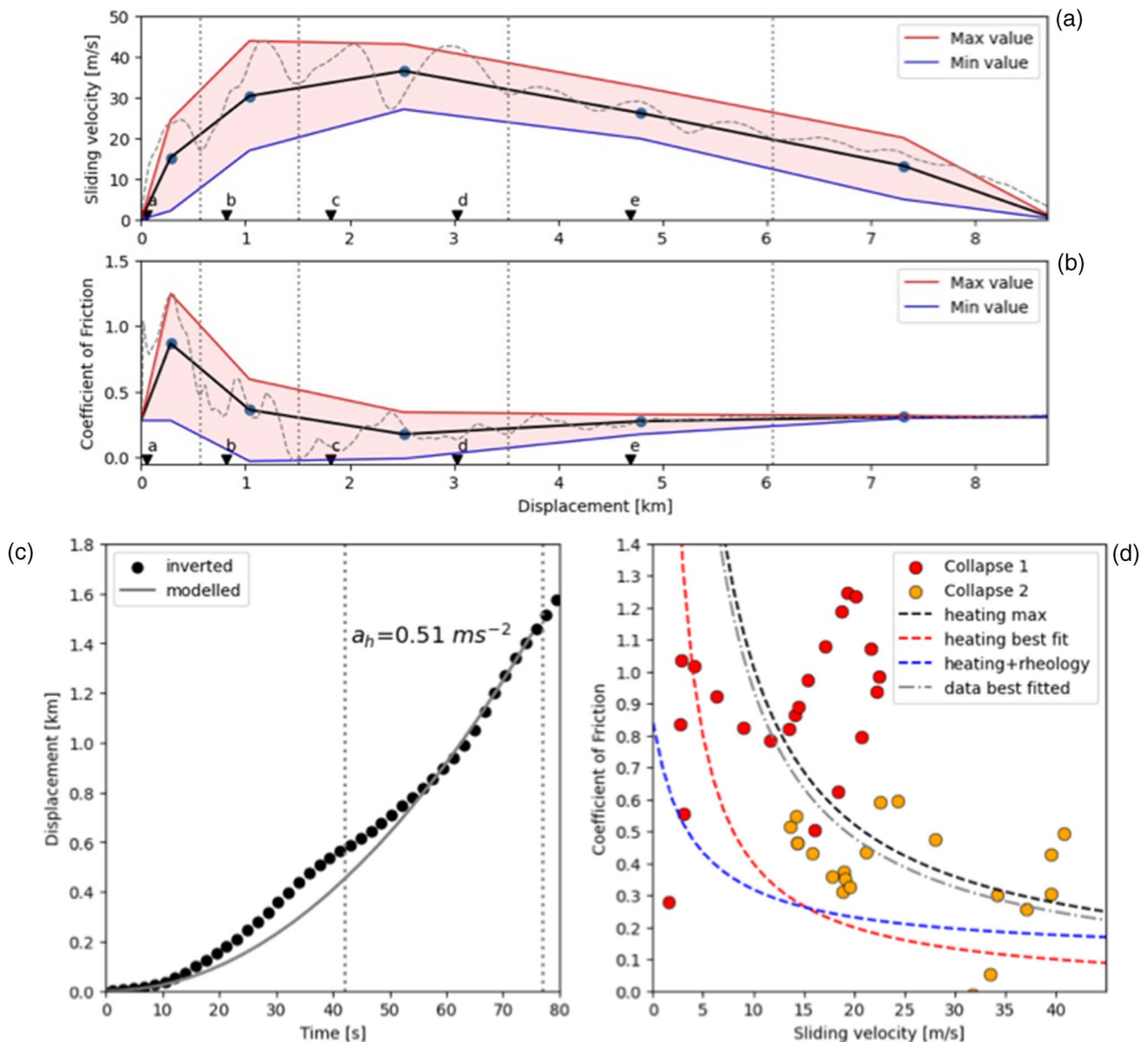


Fig. 5 Dynamic properties inferred from seismic inversion. **a** Velocity and **b** equivalent friction coefficient across the entire horizontal distance of the event. The maximum, minimum, and average value at each stage are shown. **c** Horizontal displacement throughout stages 1 and 2. The data points from inversion (dots) show a two-segment curve corresponding to the two acceleration stages, whereas the fitted line represents the equation of motion. **d** Correlation between velocity inferred from seismic signals and computed equivalent friction coefficient for the acceleration phase (Stages 1 and 2) of the 2000 Yigong landslide. Scattered data show a negative exponential relationship overlay with the empirical formula (Rice 2006; Lucas et al. 2014)

positive vertical force–time functions (Fig. 3b) that reflect downward force exerted by the bulk mass as it collapsed (Allstadt 2013). Besides, the orientation of the collapsing mass was inferred from the stronger and more complex responses in the east than the north component of the force–time functions (Fig. 3b and c), corroborating field observations made by Xin et al. (2022). This is illustrated by Point *a* in the trajectory, where the mass fell from an azimuth of 94° (Fig. 4a) with maximum force recorded throughout Stage 1 (Fig. 3d). The presence of low-frequency seismic energy (Fig. 2e, $t = 200\text{--}230$ s) with the absence of a pause in the force–time functions

between Stages 1 and 2 suggests continuous accumulated and/or accelerating collapses at near-source (Allstadt 2013).

The start time of Stage 2 is consistent between the onset of the second pulse in the 50- to 150-s vertical component of force history (Fig. 3b) and that of the second rate of change in the displacement–time graph (Fig. 5c). Between $t = 43$ and $t = 77$ s, a second phase of mass collapsed from 4490 m a.s.l. and travelled horizontally 0.95 km towards west-southwest with an average acceleration of 2.13 ms^{-2} (Fig. 5c). Point *b* marks where the sliding mass reached the Zhamu Creek (Fig. 4a), transitioning from a steep slope of 31.04°

to the gentler gradient (Fig. 4b), and achieving a maximum sliding velocity of 43.70 ms^{-1} . This impact at creek is indicated by a slight change in force direction westward (Fig. 4a), with Point *b* reflecting a strong peak in the east component and a trough in the north component of the force history (Fig. 3b). Approaching the end of Stage 2, at $t = 77 \text{ s}$, the total forces exerted by the mass reached maximum of the event marking the transition into the entrainment phase (Fig. 3b, c, and d).

The frictional strength during the acceleration was high, with equivalent coefficients of 1.24 and 0.60 in Stages 1 and 2, respectively (Fig. 5b), consistent with the experimental range of 0.6 to 0.8 for granite (Byerlee 1967; Wang et al. 2017). Fluctuations in frictional strength may represent a locking effect observed during fault rupture on rough surfaces (Ben-Zion 2001). Our interpretation benefits from the high temporal resolution of seismic observations, which offered insight to the initiation processes compared previous studies (Shang et al. 2003; Zhou et al. 2001; Wang 2008; Delaney and Evans 2015; Xin et al. 2022; Guo et al. 2023).

Entrainment: Stage 3

The mass continued to slide towards southwest along the Zhamu Creek, following a slightly sinuous path and leaving scars on the sides of the creek across Stage 3 as observed in the field (Shang et al. 2003; Yin and Xing 2012) (Fig. 4a). This behaviour likely resulted from the high potential energy gained from a high angle 1230-m drop (Fig. 4b), which converted into kinetic energy, reflected in a maximum vertical force-component of $1.92 \times 10^{11} \text{ N}$ (Fig. 3b). The sustained high kinetic energy generation may have further eroded sediments along the banks (Point *d*, Figs. 3d and 4a), potentially increasing flow mass and causing impacts against the banks (Zhang and Yin 2013; Xin et al. 2022; Gao et al. 2023a), given both velocity and force–time functions persisted for $\sim 65 \text{ s}$ (Figs. 3b–d and 5a). As the force functions peaked at $2.37 \times 10^{11} \text{ N}$ (north) and $-4.31 \times 10^{11} \text{ N}$ (east), the high sliding velocity averaged 37.8 ms^{-1} , indicating an erosion-dominant regime at $t = 87 \text{ s}$ (Point *c*) (Pudasaini and Krautblatter 2021). Meanwhile, the equivalent coefficient of friction dropped sharply from 0.58 to 0.06 (Fig. 5b), suggesting the introduction of fluid, possibly water (Brodsky et al. 2003; Singer et al. 2012). Lithological disintegration may also have contributed to this phenomenon (Taboada and Estrada 2009; Mišćević and Vlastelica 2014). The source area, consisting primarily of granite rich in feldspar and biotite (Fig. 4b), is susceptible to contrasting extreme temperature, weathering, and erosion (Huang and Li 2023; Peng et al. 2024) that weakened the frictional strength at the basal sliding plane (Singer et al. 2012; Lucas et al. 2014). This frictional weakening may signal the onset of debris flow (Fig. 5b). Dynamic properties then declined from $t = 88$ to $t = 142 \text{ s}$, with an average slope angle of 22.9° .

Transition: Stage 4

In Stage 4, as the force–time functions decrease and the process shifts from erosion to deposition, the sliding mass decelerates from a mean velocity of 36.5 to 26.1 ms^{-1} over 80 s (Fig. 5a). Topographic analysis indicates a consistent reduction in slope, fluctuating between 17.1 and 19.3° (Fig. 4b), alongside a straightened and steady runout path trending southwest down the creek (Fig. 4a). These analyses suggest that the sliding mass stabilised and moved steadily after travelling 2.57 km horizontally down the Zhamu Creek. The gentler slope with a mean of 7.4° (Fig. 4b) may have contributed to

increasing basal friction (Fig. 5b) and the observed lower velocity (Fig. 5a).

Deposition and termination: Stage 5

At $t = 262 \text{ s}$, with the prolonged deceleration, the sliding mass travelled an additional 2.73 km with a mean inclination of 6.93° (Fig. 4b) before depositing at the Yigong-Tsangpo River. Our force–time analysis and trajectory results suggest that the main sliding force during Stage 5 acted southward (Figs. 3c and 4a), and Point *e* matches exactly the elevated pile of deposits on the satellite imagery. The seismic signals returned to background noise level (Fig. 3a) and the inverted force–time functions showed minimal force exerted during deposition as they decreased to $\pm 0.2 \times 10^{11} \text{ N}$ before stabilising at $t = 485 \text{ s}$ (Fig. 3d).

Discussion

Comparison with previous studies

Our interpretations and analyses of the stages and processes of this landslide event are generally consistent with previous studies (Guo et al. 2023). We obtained a final mass of $3.09 \times 10^{11} \text{ kg}$ (Fig. 4a), approximately $1.14 \times 10^8 \text{ m}^3$ with a mass density of 2.704 gcm^{-3} based on the lithological distributions (Fig. 4b), moved by this landslide using the results from seismic inversion in this study. This is consistent with estimates from previous studies based on seismic, remote sensing and field investigations, which ranged from 2.81×10^8 to $3.06 \times 10^8 \text{ m}^3$ (Ekström and Stark 2013; Gao et al. 2023a; Guo et al. 2023). Besides, the sinuous path during Stage 3 also matched with the impact points suggested by Zhou et al. (2020) and Xin et al. (2022). Furthermore, our results are consistent with those of a recent study by Guo et al. (2024), corroborating the authors' findings regarding the entrainment area from numerical simulations, which their results give a maximum velocity of 43.6 ms^{-1} with friction coefficient dropping from 0.64 to 0.1 and highlighted the potential of friction weakening effects. Nonetheless, our findings and methodology emphasise the importance of higher-resolution time-velocity constraints, significantly enhancing near-real-time solutions for geohazard management.

However, our results indicate a longer duration and slower maximum velocity of 43.7 ms^{-1} compared to results (range 40 – 80 ms^{-1}) from numerical simulations (Zhou et al. 2020; Dai et al. 2021; Guo et al. 2023). The overall computed coefficient of friction ranges from 0.06 to 1.24 (Fig. 5b), with a steady-state value of 0.32 (Fig. S3). These readings are higher than the 0.13 reported in a laboratory study (Wang et al. 2017) and distinguishable from the static and dynamic values of 0.3 and 0.18, respectively, based on simulations and field investigation by Zhou et al. (2020) for Yigong. Furthermore, our results are significantly different from the laboratory experimental and empirical range for the friction coefficients of landslides, 0.2 to 0.8, studied using satellite imagery and field observations (Byerlee 1978; Brodsky et al. 2003; Yamada et al. 2018; Deng et al. 2020). Laboratory studies and numerical simulations often simplified the complex dynamics and neglected the boundary conditions of real-world landslides. In contrast, our high-temporal seismic-based approach, combined with field and satellite observations, provides a more comprehensive assessment of the temporal evolution of the frictional condition during the landslide event. This allows us to reveal the sub-frictionless conditions experienced

by the slide during the entrainment stage, which instead is consistent with the empirical friction coefficient of less than 0.1 for fluid flows (Huang 2022), which would have been difficult to capture using traditional laboratory or simulation techniques.

Investigating the frictional weakening

To explore the causes and mechanisms of fluidisation, we examine the avalanche dynamics concerning this unusual decline in the coefficient of friction. Specifically, we aim to understand the sharp drop in friction at the onset of Stage 3. Singer et al. (2012) reported a similarly low friction (< 0.1) in long-runout landslides in icy environments on Iapetus by studying the drop height-to-runout ratio (Heim's ratio) as a benchmark to quantify the friction coefficient using satellite images. With the glacial alpine environment in Yigong and the dynamic properties we derived from seismic inversion, we investigate the relationship between frictional strength and the velocity during the 2000 landslide. We compare our results against the friction-velocity-weakening theory proposed by Lucas et al. (2014) for Earth and Mars landslides. Our analysis focuses on the acceleration phase, where our selected seismic waveforms most accurately represent the coherent mass movement (Rice 2006; Singer et al. 2012; Allstadt 2013; Ekström and Stark 2013; Lucas et al. 2014) (Figs. 5d and S3). Our data show the best fit of a negative exponential relationship between the velocity and frictional strength of the sliding mass. Our data reveal a negative exponential correlation between velocity and frictional strength, showing a clear trend of decreasing friction with increasing velocity, and vice versa. This phenomenon corresponds to the frictional velocity weakening (Lucas et al. 2014) and the flash heating theory (Rice 2006). The best-fit curve aligns more closely with flash heating theory rather than friction-velocity weakening, suggesting flash heating occurred during the acceleration phase.

The geographical and environmental context of Yigong plays a critical role in flash heating, which caused the significant friction reduction during the 2000 landslide. Flash heating occurs predominantly when two coherent blocks shear, generating basal friction that converts excess energy into heat (Goren and Aharonov 2007). In this case, the rock-glacier collapse observed in Stages 1 and 2 created a large surface area of 1.4 km² (Dai et al. 2021) and a 1,230 m drop height, producing substantial friction and converting excessive potential and kinetic energy into heat, favouring partial melting along the sliding plane of a glacier in such an alpine environment (Singer et al. 2012). Despite significant mass erosion, the thin sediment layer deposited at the creek's base suggests fluidisation of the sliding mass (Tomczyk 2021; Gao et al. 2023a; Bhuyan et al. 2024), where melted glacier ice mixed with sediments.

Additionally, the length of the Zhamu Creek offered an extended path for the mass to travel (Tomczyk 2021; Gao et al. 2023a; Bhuyan et al. 2024). This enhanced sediment erosion and mass accumulation before stopping at the Yigong-Tsangpo River. The larger the avalanche, the greater the sliding surface area, leading to even stronger friction at high speed, hence more prone to heat generation (Lucas et al. 2014; Deng et al. 2020). The partially melted water further increased the pore pressure in the slumped sliding mass around Spring when this event happened, hence reducing the frictional strength sharply (Goren and Aharonov 2007), allowing the

velocity to remain at a high value of 43.18 to 43.65 ms⁻¹ during stage 3.

Limitations

This study employs high temporal resolution seismic data to analyse the dynamic properties of the landslide. But the lack of data availability and insufficient robust case studies for comparison necessitates several assumptions, thus we can only assess the possibility of flash heating and friction-velocity weakening theory to explain the sharp frictional reduction phenomenon observed in Fig. 5b. While it identifies factors influencing friction, additional data and case investigations are needed to better understand the other elements controlling landslide processes, including but not limited to temperature, fluid pressure, cohesion of particles, fragmentation spreading, and grain size (Vardoulakis 2002). To assess other mechanisms, we also need auxiliary data to support the investigation; for instance, terrain data for thickness to examine the rheology of flow (Lucas et al. 2014).

In addition, there are limitations in using seismic data and inversion to infer dynamic properties due to simplification of our methodology, such as the assumption of a simple block model and the inability to fully represent the transition to an incoherent debris flow (Allstadt 2013; Ekström and Stark 2013; Lucas et al. 2014). These should be considered in future analyses, particularly resolving for the growing volume of sliding mass throughout the landslide. For simplicity, we assumed a constant average mass in the model as a constraint for the dynamics. Although this assumption leads to an overestimation of mass in Stages 1 and 2, the accelerations will be larger in the two stages with potentially smaller mass, which will further strengthen our result that the collapsing bulk mass accelerated during these two stages. Moreover, this seismic method only accounted for a coherent bulk mass, hence, motion inferred beyond the acceleration stage does not closely align with an incoherent debris flow that will elongate and alter its shape as it flows down the Zhamu Creek. More complexity is also expected during entrainment and transition. Employing high-frequency seismic analysis and Empirical Green's Function inversion (Zhang et al. 2020; Zhang et al. 2021a, b) may provide more insights into the complex flow dynamics during these phases.

Besides, due to the limitations of our current observations and the constraints in our methodology, there might be uncertainties in explaining the initiation mechanisms. We proposed that rocks and glaciers gradually collapsed in a series of disintegrations, triggering further instability first in the east and then the north of the cliff during Stages 1 and 2, leading to a volume build-up (Xin et al. 2022) (Fig. 4a). This interpretation is supported by fluctuations (sub-peaks) in the east and vertical components of the force-time function during the 10- to 150-s analysis (Fig. 3b and d), likely indicating smaller-scale collapses (Allstadt 2013). The increasing force magnitudes across all components during Stages 1 and 2 (Fig. 3b–d) suggest acceleration of the collapsing mass. Alternatively, the rising seismic amplitudes (Fig. 3a) may indicate a growing volume released in two stages. Our preferred interpretation aligns with previous studies (Xu et al. 2012; Xin et al. 2022) and suggests a two-collapse mechanism, with the main collapse directed westward, leaving a V-shaped scar at the source (Fig. 1d).

Conclusion

We analysed the force history of the 2000 Yigong landslide through seismic inversion combined with observations from satellite image study and field survey. This revealed the complexities in the dynamic properties of the event through five stages, and the high temporal resolution allowed us to reveal the friction evolution throughout the event. We further investigated the possible factors that caused an abrupt drop in basal friction that potentially led to the long-runout characteristic of the landslide. We find that the 2000 Yigong landslide began at 11:58:56 UTC on 9th April, lasted for 485 s, and transported a vast mass of 3.09×10^{11} kg (1.14×10^8 m³ with densities of 2.704 gcm⁻³) throughout five stages. Our results indicated two collapses during the acceleration phase in the avalanche, each stage with rock-glacier dis-integrations. Upon the entrainment phase, the mass reached its maximum velocity of 43.7 ms⁻¹ and met the Zhamu Creek with maximum total forces of 5.0×10^{11} N exerted on the ground. Notably, we calculated the equivalent friction coefficient throughout the landslide for the first time, benefiting from the high temporal resolution of seismic data. Our friction analyses suggested that, at around $t = 77$ s, there was enormous heat generated from the excessive basal friction during sliding in Stages 1 and 2, which partially melted the collapsed glacier. This fluidised the mass to a debris flow to run down the Zhamu Creek with high mobility, resulting in a long runout distance. The mass eventually decelerated and deposited at the Yigong-Tsangpo River.

Acknowledgements

The authors express their gratitude to Siming He, Lei Zhu, and Huanhuan Zhang from the Chinese Academy of Sciences for their insightful comments and discussions regarding landslide basal friction.

Funding

This research is funded by The Chinese University of Hong Kong under Grant Numbers 4053594, 4200720, and 3135043, and the Croucher Tak Wah Mak Innovation Award.

Data availability

All data used in this study are openly accessible. Waveform data are provided by the Data Management Centre of the Incorporated Research Institutions for Seismology (IRIS) Synthetics Engine web service (<https://ds.iris.edu/gmap/>). Topographic data are sourced from the GMRT DEM available at <https://www.gmrt.org>. Satellite imagery is obtained from Landsat 7 and extracted via Google Earth Engine (<https://earthengine.google.com>). The US Geological Survey Software Release, *lsforce* (Version 1.0.1), along with source code by Toney and Allstadt (2021), is accessible at <https://code.usgs.gov/ghsc/lhp/lsforce>.

Declarations

Competing interests The authors declare no competing interests.

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References

- Aki K, Richards PG (1980) Quantitative seismology: theory and methods. W.H. Freeman, San Francisco
- Allstadt K (2013) Extracting source characteristics and dynamics of the August 2010 Mount Meager landslide from broadband seismograms. *J Geophys Res Earth Surf* 118:1472–1490. <https://doi.org/10.1002/jgrf.20110>
- Ben-Zion Y (2001) Dynamic ruptures in recent models of earthquake faults. *J Mech Phys Solids* 49(9):2209–2244. [https://doi.org/10.1016/S0022-5096\(01\)00036-9](https://doi.org/10.1016/S0022-5096(01)00036-9)
- Bhuyan K, Rana K, Ferrer JV, Cotton F, Ozturk U, Catani F, Malik N (2024) Landslide topology uncovers failure movements. *Nat Commun* 15:2633. <https://doi.org/10.1038/s41467-024-46741-7>
- Brodsky EE, Gordeev E, Kanamori H (2003) Landslide basal friction as measured by seismic waves. *Geophys Res Lett* 30(24):2236. <https://doi.org/10.1029/2003GL018485>
- Byerlee JD (1967) Frictional characteristics of granite under high confining pressure. *J Geophys Res* 72(14):3639–3648. <https://doi.org/10.1029/JZ072i014p03639>
- Byerlee JD (1978) Friction of rocks. *PAGEOPH* 116:615–626. <https://doi.org/10.1007/BF00876528>
- Castelli M, Scavia C, Bonnard C, Laloui L (2009) Mechanics and velocity of large landslides. *Eng Geol* 109(1–2):1–4. <https://doi.org/10.1016/j.enggeo.2009.07.009>
- Dai Z, Xu K, Wang F, Yang H, Qin S (2021) Numerical investigation on the kinetic characteristics of the Yigong debris flow in Tibet, China. *Water* 13:1076. <https://doi.org/10.3390/w13081076>
- Delaney KB, Evans SG (2015) The 2000 Yigong landslide (Tibetan Plateau), rockslide-dammed lake and outburst flood: review, remote sensing analysis, and process modelling. *Geomorphology* 246:377–393. <https://doi.org/10.1016/j.geomorph.2015.06.020>. (ISSN 0169-555X)
- Deng Y, Yan S, Scaringi G, Liu W, He S (2020) An empirical power density-based friction law and its implications for coherent landslide mobility. *Geophys Res Lett* 47(11). <https://doi.org/10.1029/2020GL087581>
- Deparis J, Jongmans D, Cotton F, Baillet L, Thouvenot F, Hantz D (2008) Analysis of rock-fall and rock-fall avalanche seismograms in the French Alps. *Bull Seismol Soc Am* 98:1781–1796. <https://doi.org/10.1785/0120070082>
- Ekström G, Stark CP (2013) Simple scaling of catastrophic landslide dynamics. *Science* 339(6126):1416–1419. <https://doi.org/10.1126/science.1232887>
- Fukao Y (1995) Single-force representation of earthquakes due to landslides or the collapse of caverns. *Geophys J Int* 122:243–248. <https://doi.org/10.1111/j.1365-246X.1995.tb03551.x>
- Gao H, Gao Y, Li B, Yin Y, Yang C, Wan J, Zhang T (2023a) The dynamic simulation and potential hazards analysis of the Yigong landslide in Tibet, China. *Remote Sens* 15(5):1322. <https://doi.org/10.3390/rs15051322>
- Gao Y, Li B, Gao H, Gao S, Wang M, Liu X (2023b) Risk assessment of the Sedongpu high-altitude and ultra-long-runout landslide in the lower Yarlung Zangbo River, China. *Bull Eng Geol Environ* 82:360. <https://doi.org/10.1007/s10064-023-03374-2>
- Goren L, Aharonov E (2007) Long runout landslides: the role of frictional heating and hydraulic diffusivity. *Geophys Res Lett* 34:L07301. <https://doi.org/10.1029/2006GL028895>
- Guo C, Montgomery DR, Zhang Y, Zhong N, Fan C, Wu R, Yang Z, Ding Y, Jin J, Yan Y (2020) Evidence for repeated failure of the giant Yigong landslide on the edge of the Tibetan Plateau. *Sci Rep* 10:14371. <https://doi.org/10.1038/s41598-020-71335-w>

- Guo C, Yuan H, Wu R, Yan Y, Yang Z (2023) Research review and prospects of the 2000 giant Yigong long-runout landslide: volume, formation mechanism and recurrence period, Tibetan plateau, China. *Front Earth Sci* 10. <https://doi.org/10.3389/feart.2022.1017611>
- Guo Z, Zhou X, Huang D, Zhai S, Tian B, Li G (2024) Dynamic simulation insights into friction weakening effect on rapid long-runout landslides: a case study of the Yigong landslide in the Tibetan Plateau, China. *China Geol* 7(2):222–236. <https://doi.org/10.31035/cg2023132>. (ISSN 2096-5192)
- Handwerger AL, Rempel AW, Skarbek RM, Roering JJ, Hilley GE (2016) Rate-weakening friction characterizes both slow sliding and catastrophic failure of landslides. *Proc Natl Acad Sci* 113:10281–10286. <https://doi.org/10.1073/pnas.1607009113>
- Handwerger AL, Huang MH, Fielding EJ, Booth A, Bürgmann R (2019) A shift from drought to extreme rainfall drives a stable landslide to catastrophic failure. *Sci Rep* 9:1569. <https://doi.org/10.1038/s41598-018-38300-0>
- Hibert C, Ekström G, Stark CP (2014) Dynamics of the Bingham Canyon Mine landslides from seismic signal analysis. *Geophys Res Lett* 41:4535–4541. <https://doi.org/10.1002/2014GL060592>
- Hibert C, Ekström G, Stark CP (2017) The relationship between bulk-mass momentum and short-period seismic radiation in catastrophic landslides. *J Geophys Res Earth Surf* 122:1201–1215. <https://doi.org/10.1002/2016JF004027>
- Huang S (2022) Reading the Moody chart with a linear interpolation method. *Sci Rep* 12:6587. <https://doi.org/10.1038/s41598-022-10552-x>
- Huang Y, Li T (2023) On the relationship between the mechanical strength and mineral microstructure of biotite granite under temperature influence. *Front Earth Sci* 11:1209433. <https://doi.org/10.3389/feart.2023.1209433>
- Huggel C, Clague J, Korup O (2012) Is climate change responsible for changing landslide activity in high mountains? *Earth Surf Process Landf* 37:77–91. <https://doi.org/10.1002/esp.223>
- Kanamori H, Given JW (1982) Analysis of long-period seismic waves excited by the May 18 1980, eruption of Mount St. Helens—a terrestrial monopole? *J Geophys Res* 87(B7):5422–5432. <https://doi.org/10.1029/JB087iB07p05422>
- Kennett BLN (1991) IASPEI 1991 Seismological tables. Bibliotech, Canberra, Australia, 167 pp. <https://doi.org/10.1111/j.1365-3121.1991.tb00863.x>
- Krischer L, Hutko A, Driel M, Stähler S, Bahavar M, Trabant C, Nissen-Meyer T (2017) On-demand custom broadband synthetic seismograms. *Seismol Res Lett* 88. <https://doi.org/10.1785/0220160210>
- Lee H, Chung S, Wang J, Wen D, Lo C, Yang TF, Zhang Y, Xie Y, Lee T, Wu G, Ji J (2003) Miocene Jiali faulting and its implications for Tibetan tectonic evolution. *Earth Planet Sci Lett* 205(3–4):185–194. [https://doi.org/10.1016/S0012-821X\(02\)01040-3](https://doi.org/10.1016/S0012-821X(02)01040-3)
- Li W, Chen Y, Liu F, Yang H, Liu J, Fu B (2019) Chain-style landslide hazardous process: constraints from seismic signals analysis of the 2017 Xinmo landslide, SW China. *J Geophys Res: Solid Earth* 124:2025–2037. <https://doi.org/10.1029/2018JB016433>
- Lin CH (2015) Insight into landslide kinematics from a broadband seismic network. *Earth Planet Sp* 67:8. <https://doi.org/10.1186/s40623-014-0177-8>
- Lin CH, Kumagai H, Ando M, Shin TC (2010) Detection of landslides and submarine slumps using broadband seismic networks. *Geophys Res Lett* 37:L22309. <https://doi.org/10.1029/2010GL044685>
- Lucas A, Mangeney A, Ampuero JP (2014) Frictional velocity-weakening in landslides on Earth and on other planetary bodies. *Nat Commun* 5:3417. <https://doi.org/10.1038/ncomms4417>
- Miščević P, Vlastelica G (2014) Impact of weathering on slope stability in soft rock mass. *J Rock Mech Geotech Eng* 6(3):240–250. <https://doi.org/10.1016/j.jrmge.2014.03.006>. (ISSN 1674-7755)
- Morino C, Conway SJ, Balme MR, Helgason JK, Sæmundsson Þ, Jordan C, Hillier J, Argles T (2021) The impact of ground-ice thaw on landslide geomorphology and dynamics: two case studies in northern Iceland. *Landslides* 2785–2812. <https://doi.org/10.1007/s10346-021-01661-1>
- Mukhopadhyay B, Dasgupta S (2015) Earthquake swarms near eastern Himalayan Syntaxis along Jiali Fault in Tibet: a seismotectonic appraisal. *Geosci Front* 6(5):715–722. <https://doi.org/10.1016/j.gsf.2014.12.009>
- Peng L, Li X, Peng X, Gan Y, Wang J (2024) Analysis of physical and mechanical behaviors and microscopic mineral characteristics of thermally damaged granite. *Sci Rep* 14:14776. <https://doi.org/10.1038/s41598-024-65752-4>
- Pudasaini SP, Krautblatter M (2021) The mechanics of landslide mobility with erosion. *Abstr Nat Commun* 12(1). <https://doi.org/10.1038/s41467-021-26959-5>
- Rice RJ (2006) Heating and weakening of faults during earthquake slip. *J Geophys Res* 111:B05311
- Shang Y, Yang Z, Li L, Liu D, Liao Q, Wang Y (2003) A super-large landslide in Tibet in 2000: background, occurrence, disaster, and origin. *Geomorphology* 54(3–4):225–243. [https://doi.org/10.1016/S0169-555X\(02\)00358-6](https://doi.org/10.1016/S0169-555X(02)00358-6)
- Singer KN, McKinnon WB, Schenectady PM, Moore JM (2012) Massive ice avalanches on lapetus mobilised by friction reduction during flash heating. *Nat Geosci* 5:574–578. <https://doi.org/10.1038/ngeo1526>
- Suriñach E, Vilajosa I, Khazaradze G, Biescas B, Furdada G, Vilaplana JM (2005) Seismic detection and characterization of landslides and other mass movements. *Nat Hazards Earth Syst Sci* 5:791–798. <https://doi.org/10.5194/nhess-5-791-2005>
- Taboada A, Estrada N (2009) Rock-and-soil avalanches: theory and simulation. *J Geophys Res* 114:F03004. <https://doi.org/10.1029/2008JF001072>
- Takei Y, Kumazawa M (1994) Why have the single force and torque been excluded from seismic source models? *Geophys J Int* 118(1):20–30. <https://doi.org/10.1111/j.1365-246X.1994.tb04672.x>
- Tomczyk AM (2021) Morphometry and morphology of fan-shaped landforms in the high-Arctic settings of central Spitsbergen, Svalbard. *Geomorphology* 392:107899. <https://doi.org/10.1016/j.geomorph.2021.107899>
- Toney L, Allstadt KE (2021) Isforce: A Python-based single-force seismic inversion framework for massive landslides. *Seismol Res Lett* 92(4):2610–2626. <https://doi.org/10.1785/0220210004>
- Vardoulakis I (2002) Dynamic thermo-pop-mechanical analysis of catastrophic landslides. *Geotechnique* 52(3):157–171
- Wang ZH (2008) A thunder at the beginning of the 21st century-the giant Yigong landslide In: The 10th international symposium on landslides and engineered slopes, pp 1068–1075
- Wang YF, Dong JJ, Cheng QG (2017) Velocity-dependent frictional weakening of large rock avalanche basal facies: implications for rock avalanche hypermobility? *J Geophys Res* 122(3):1555–2430. <https://doi.org/10.1002/2016JB013624>
- Xin P, Wang T, Liu J, Liu F, Du J, Zhao J (2022) The geological structure and sliding mode of the slopes in the Yigong landslide source area, Tibet. *J Geomech* 28(6):1012–1023. <https://doi.org/10.12090/j.issn.1006-6616.2022072>
- Xu Q, Shang Y, van Asch T, Wang S, Zhang Z, Dong X (2012) Observations from the large, rapid Yigong rock slide—debris avalanche, southeast Tibet. *Can Geotech J* 49(5):589–606. <https://doi.org/10.1139/t2012-021>
- Yamada M, Matsushi Y, Chigira M, Mori J (2012) Seismic recordings of landslides caused by Typhoon Talas (2011), Japan. *Geophys Res Lett* 39:L13301. <https://doi.org/10.1029/2012GL052174>
- Yamada M, Mangeney A, Matsushi Y, Matsuzawa T (2018) Estimation of dynamic friction and movement history. *Landslides* 15:1963–1974. <https://doi.org/10.1007/s10346-018-1002-4>
- Yin Y, Xing A (2012) Aerodynamic modeling of the Yigong gigantic rock slide-debris avalanche Tibet China. *Bull Eng Geol Environ* 71(1):149–160. <https://doi.org/10.1007/s10064-011-0348-9>
- Zhang M, Yin Y (2013) Dynamics. Mobility-controlling factors and transport mechanisms of rapid long-runout rock avalanches in China. *Eng Geol* 167:37–58. <https://doi.org/10.1016/j.enggeo.2013.10.010>

- Zhang Z, He S, Liu W, Liang H, Yan S, Deng Y, Bai X, Chen Z (2019) Source characteristics and dynamics of the October 2018 Baige landslide revealed by broadband seismograms. *Landslides* 16:777–785. <https://doi.org/10.1007/s10346-019-01145-3>
- Zhang Z, He S, Li Q (2020) Analyzing high-frequency seismic signals generated during a landslide using source discrepancies between two landslides. *Eng Geol* 272:105640. <https://doi.org/10.1016/j.enggeo.2020.105640>
- Zhang Z, Walter F, McArdell BW, Wenner M, Chmiel M, de Haas T, He S (2021a) Insights from the particle impact model into the high-frequency seismic signature of debris flows. *Geophys Res Lett* 48(1). <https://doi.org/10.1029/2020GL088994>
- Zhang Z, Walter F, McArdell BW, de Haas T, Wenner M, Chmiel M, He S (2021b) Analyzing bulk flow characteristics of debris flows using their high frequency seismic signature abstract plain language summary key points. *J Geophys Res Solid Earth* 126(12). <https://doi.org/10.1029/2021JB022755>
- Zhang Z, Liu M, Tan YJ, Walter F, He S, Chmiel M, Su J (2024a) Landslide hazard cascades can trigger earthquakes. *Nat Commun* 15:2878. <https://doi.org/10.1038/s41467-024-47130-w>
- Zhang Z, Tan YJ, Walter F, He S, Chmiel M, Su J (2024b) Seismic monitoring and geomorphic impacts of the catastrophic 2018 Baige landslide hazard cascades in the Tibetan plateau. *J Geophys Res: Earth Surf* 129:e2023JF007363. <https://doi.org/10.1029/2023JF007363>

- Zhou CH, Yue ZQ, Lee CF, Zhu BQ, Wang ZH (2001) Satellite image analysis of a huge landslide at Yi Gong, Tibet, China. *Q J Eng GeolHydrogeol* 34:325–332. <https://doi.org/10.1144/qjengh.34.4.325>
- Zhou GG, Roque PJC, Xie Y, Song D, Zou Q, Chen H (2020) Numerical study on the evolution process of a geohazards chain resulting from the Yigong landslide. *Landslides* 17(11):2563–2576. <https://doi.org/10.1007/s10346-020-01448-w>

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Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10346-025-02525-8>.

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